GATE-AII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS

Lecture No.- 05



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Recap of Previous Lecture



Topic Existence of limits and how to evaluate limits

Topics to be Covered



Topic How to evaluate the limits?

Topic Concept of continuity

Topic Questions based on continuity

$$L = \lim_{x \to a} [f(x)]^{g(x)} = \lim_{x \to a} [\text{function}]^{\text{function}}$$

Strategy : Taking Log both sides

general strategy

$$\ln L = \lim_{x \to a} g(x) \ln [f(x)]$$

$$L = e^{\lim_{x \to a} g(x) \ln [f(x)]}$$

$\to \infty^{\to \infty}$
$\to 1^{\to \infty}$
$\to 0^{\to \infty}$

$(\to \infty)^{\to \infty}$ $(\to 0)^{\to \infty}$
$(\to 1)^{\to \infty}$

Indeterminate forms

$\frac{\to 0}{\to 0}$ $\frac{\to \infty}{\to \infty}$
$\to \infty - \infty$

special case $\longrightarrow (\to 1)^{\to \infty}$

$$L = \lim_{x \to a} [f(x)]^{g(x)}$$

Where $A = \lim_{x \to a} [f(x) - 1]\, g(x)$

$\text{Ans} = e^{A}$

$\to 1^{\to \infty}$

1) $\ln L = \lim_{x \to a} g(x) \ln f(x)$ ⟶ STRATEGY

2) $\to 1^{\to \infty}$ form ⟶ e^{A}

$[f(x)]^{g(x)}$ Where $\boxed{A = \lim_{x \to a} [f(x) - 1]\, g(x)}$

Convert to The L-Hospital Rule

$$\lim_{x \to a} g(x) \ln f(x)$$

$$= \lim_{x \to a} \frac{\ln f(x)}{\frac{1}{g(x)}} = \text{L-Hospital Rule} \quad \frac{\to 0}{\to 0} \text{ OR } \frac{\to \infty}{\to \infty}$$



Topic : Single Variable Calculus

#Q. Evaluate : $\lim_{x\to 0}(1-2x)^{1/x}$

$$\lim_{x\to 0}[1-2x]^{\frac{1}{x}}$$

Type $\to (\to 1)^{\to \infty}$ form

Taking ln both sides

$$\ln L = \lim_{x\to 0}\frac{1}{x}\ln(1-2x) = \lim_{x\to 0}\frac{\ln(1-2x)}{x} = \left(\frac{\to 0}{\to 0}\right) \text{form}$$

apply L-Hospital Rule

$$= \lim_{x\to 0}\frac{\frac{d}{dx}\ln(1-2x)}{\frac{d}{dx}(x)} = \lim_{x\to 0}\frac{\frac{1}{(1-2x)}\cdot(-2)}{1}$$

$$= \lim_{x\to 0}\frac{-2}{(1-2x)}$$

Free form
Indeterminate forms

Plug in Limits $\Rightarrow -2$

$\ln L = -2 \Rightarrow \boxed{L = e^{-2}}$ → Ans

$$\lim_{x\to 0}(1-2x)^{\frac{1}{x}} \longrightarrow (\to 1)^{\to\infty}$$

$f(x)=1-2x \quad g(x)=\frac{1}{x}$

$= e^{A}$

Where $A = \lim_{x\to 0}[f(x)-1]\,g(x)$

$= \lim_{x\to 0}[(1-2x)-1]\cdot\frac{1}{x}$

$= \lim_{x\to 0}\frac{-2x}{x}$

$= -2$

$= e^{-2}$ Ans

ONLY $(\to 1^{\to\infty})$



Topic : Single Variable Calculus

$L = e^{\frac{1}{\pi}}$

3-5 minute

#Q. Evaluate : $\lim_{x \to 1} x^{\cot \pi x}$

$(1)^{\to \infty}$ (2) Taking ln both sides

$= e^{A}$

$\tan(\pi - \theta) = -\tan\theta$

Where $A = \lim_{x \to 1} [f(x) - 1]\, g(x)$

$= \lim_{x \to 1} [(x-1)] \cot \pi x$

Let $\lim_{x \to 1} \left[\frac{(\pi - \pi x)}{\tan(\pi - \pi x)}\right]^{\frac{1}{\pi}}$

Template

$= \lim_{x \to 1} \frac{(x-1)}{\tan \pi x}$ = Template

Multiply via π and divide also

$= \lim_{x \to 1} \frac{(\pi x - \pi)}{-\tan(\pi - \pi x)} \frac{1}{\pi}$

$= \lim_{x \to 1} \frac{+(\pi - \pi x)}{+\tan(\pi - \pi x)} \frac{1}{\pi}$

$= \frac{1}{\pi}$

$L = e^{\frac{1}{\pi}}$

Ans

$\lim_{x \to 0} \frac{\tan x}{x} = 1 \quad \lim_{x \to 0} \frac{\tan f(x)}{f(x)} = 1$

$$\lim_{x\to 1} \frac{(\pi-\pi x)}{\tan(\pi-\pi x)} \cdot \frac{1}{\pi}$$

$x \to 1 \quad \pi-\pi x \longrightarrow 0$

$f(x) \longrightarrow 0$

$$\lim_{x\to 1} \frac{(\pi-\pi x)}{\tan(\pi-\pi x)} \left(\frac{1}{\pi}\right)$$

Template

$$\lim_{x\to 1} \frac{f(x)}{\tan[f(x)]}$$

$f(x) \to 0$



Topic : Single Variable Calculus

13-15 / 13] Just do It

#Q. The value of $\lim\limits_{x\to a}\left(2-\frac{a}{x}\right)^{\tan\frac{\pi x}{2a}}$ is:

[1^{∞} Type of indeterminate form]

A e

B e^{π}

C $e^{-2/\pi}$

D $e^{2/\pi}$

$$= \lim_{x\to a}\left[\frac{+\frac{a}{x^2}}{-\operatorname{cosec}^2\frac{\pi x}{2a}\cdot\frac{\pi}{2a}}\right]$$

$$= -\frac{2}{\pi}$$

$$L = e^{-2/\pi}$$

$$\lim_{x\to a}\left(2-\frac{a}{x}\right)^{\tan\frac{\pi x}{2a}}$$

$\to \infty$

$(\to 1)^{[f(x)-1]\,g(x)}$

Where $A = \lim\limits_{x\to a}\left(2-\frac{a}{x}-1\right)\tan\frac{\pi x}{2a}$

$$= \lim_{x\to a}\frac{\left[1-\frac{a}{x}\right]}{\cot\frac{\pi x}{2a}} = \frac{\to 0}{\to 0}$$

Using L-Hospital Rule

$$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$



Topic : Single Variable Calculus

#Q. The value of L= $\lim_{x \to 0} (1/x)^{\sin x}$ is:

[∞^0 Type of indeterminate form]

A 0

B 1/2

C 1

D None of these

$$L = \lim_{x\to 0} \left(\frac{1}{x}\right)^{\sin x}$$

$(\to\infty)^{\to 0}$ → Infinity is Not a No

Taking ln both sides

$(\to 1)^{\to\infty}$ ✓

$$\ln L = \lim_{x\to 0} \sin x \ln\left(\frac{1}{x}\right)$$

$$= \lim_{x\to 0} \frac{\ln\left(\frac{1}{x}\right)}{\frac{1}{\sin x}} \quad \frac{\to\infty}{\to\infty} \text{ form}$$

Using L-Hospital Rule

$$= \lim_{x\to 0} \frac{\ln\left(\frac{1}{x}\right)}{\frac{1}{\sin x}} = \lim_{x\to 0} \frac{+\frac{1}{x}}{+\frac{\cos x}{\sin^2 x}}$$

$$= \lim_{x\to 0} \frac{\sin^2 x}{x\cos x}$$

Plug in $= 0$

$$\frac{d}{dx}\left(\frac{1}{\sin x}\right) = -\frac{1}{\sin^2 x}\cos x$$

$$\frac{d}{dx}\left[\frac{\ln\left(\frac{1}{x}\right)}{x}\right] = \frac{1}{\frac{1}{x}}\cdot\left(\frac{-1}{x^2}\right)$$

$$= -x\cdot\frac{1}{\left(\frac{x^2}{x}\right)} = \boxed{-\frac{1}{x}}$$

✗ first mistake

$\vec{\infty} + \vec{\infty} = 2\vec{\infty}$

$\frac{\vec{\infty}}{\to\infty} = 1$ Wrong

$\vec{\infty} \times \vec{\infty} = \vec{\infty}^2$

$\vec{\infty} - \vec{\infty} = 0$

→ Again L-Hospital Rule

$\ln L = 0$

$\boxed{L = e^0 = 1}$

Ques –

Calculate the Limit $L = \lim_{x\to\infty} (1+x^2)^{e^{-x}}$

$$L = \lim_{x\to\infty} (1+x^2)^{e^{-x}}$$

Taking log both sides

$$\log L = \lim_{x\to\infty} e^{-x} \log(1+x^2)$$

$$= \lim_{x\to\infty} \frac{\log(1+x^2)}{e^{+x}} = \frac{\to\infty}{\to\infty}$$

$$= \lim_{x\to\infty} \frac{\frac{1}{(1+x^2)}\,2x}{e^x}$$

$$= \lim_{x\to\infty} \frac{2x}{(1+x^2)e^x} = \lim_{x\to\infty} \frac{2}{e^x + x^2e^x + 2xe^x} = 0$$

remove Indeterminate



apply L-Hospital Rule

$$L = e^0$$

$$L = 1$$ Ans



Topic : Single Variable Calculus

GATE

#Q. Evaluate :

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{2x}$$ is equal to

$$L=\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{2x} \quad (\hookrightarrow 1^{\infty})$$

$$A=\lim_{x\to\infty}\left[1+\frac{1}{x}-1\right]^{2x}$$

(A) – declared Answer

$$=2$$

$$L=e^2$$

A e^2

B e

C 1

D e



2 mins Summary

- Topic One
- Topic Two
- Topic Three
- Topic Four
- Topic Five

$\lim_{x \to a} [f(x)]^{f(x)} = \begin{vmatrix} (\overrightarrow{\infty})^{\to \infty} \\ (\overrightarrow{0})^{\to \infty} \end{vmatrix}$

$(\to 1^{\to \infty})$ SHORTcut $\to e^{A}$

Questions

Rahul Sir PW { # Reverse Limit concept
continuity

THANK - YOU